

- 1 In this question the number of points on a driver's licence for motoring offences is denoted by n .

When calculating a driver's premium, a particular insurance company takes account of such offences by multiplying the premium by an amount M , where

$$M = 1 \quad \text{if } n \leq 3,$$

$$M = 2^{\frac{n}{6}} \quad \text{if } 3 < n < 12.$$

A driver who is insured with this company is paying a premium of £520. Initially he has no points on his licence.

He is convicted of a speeding offence and so receives 3 points on his licence. He is then convicted for dangerous driving and receives a further 6 points on his licence.

Calculate his premium

(A) after his first offence,

(B) after his second offence.

[2]

5(A)	
5(B)	

2 In lines 68 to 69, the article says

“Clearly the role of inexperience goes down with age. A possible mathematical model is that it decays exponentially.”

This question investigates this in the case of male drivers. The relevant part of Table 4 is reproduced here.

Age (in years)	18.0	20.5	23.0	27.5	45.0
Male %	36	15	6	2.5	1

A model is proposed in which

$$y = a + be^{-k(x-17)},$$

where

- x is the age of the driver,
- y is the percentage of accidents in which inexperience played a role,
- a , b and k are constants to be determined.

(i) Draw a sketch graph of y against x for values of x between 18 and 45.

[1]

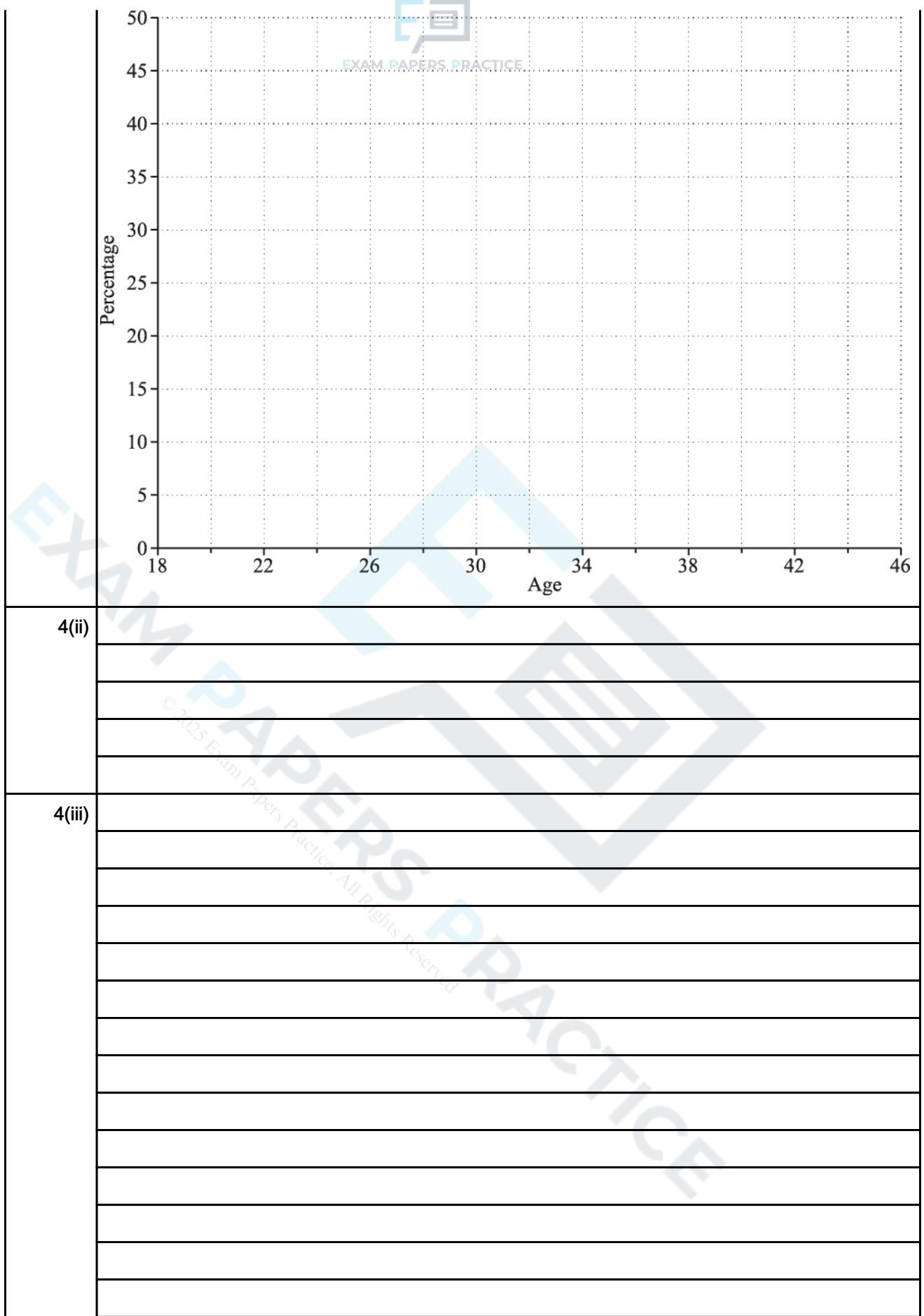
(ii) Explain why the graph justifies putting $a = 1$.

[1]

(iii) Taking $a = 1$ and using the first two data points, it is estimated that $b = 50.5$ and $k = 0.3665$. Investigate whether these values are consistent with the point (23, 6).

[2]

4(i)	
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- 3 On June 1st 2007 Louise paid her first car insurance premium. She did not have any no-claims discount.

She retained her policy with the same insurance company. She had no accidents and so did not make any claim on her insurance. Her no-claims discount followed the pattern in Table 7. Apart from her no-claims discount, her basic premium remained the same.

On May 31st 2014, Louise calculated that so far her no-claims discount had saved her £3875.

What premium did Louise pay on June 1st 2007?

[3]

3	



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4 In lines 97 to 100, the article says

“Most insurance companies have a maximum no-claims discount of 65%. One way of interpreting this practice is that the figure arrived at by applying the maximum no-claims discount is actually the basic cost of the insurance, and that drivers who have not earned the so-called discount are actually paying a surcharge.”

A new driver without any no-claims discount pays k times “the basic cost of the insurance”.

Find the value of k .

[2]

2	

- 5 The example illustrated in Table 1 in the article referred to the members of a racing club.

This question is about a similar club and the equivalent table is given below. The club has 150 members and each of them belongs to one of the groups P, Q, R or S.

For each group the probability of an accident and the average cost is given in the table below.

Group	P	Q	R	S
Proportion of people	10%	20%	40%	30%
Probability of an accident	0.1	0.15	0.05	0.2
Average cost per accident	£5000	£2000	£1000	£500

- (i) Complete the table below.

[3]

- (ii) Calculate the premium if each member of the club pays the same amount and the total of the premiums is equal to the total average cost of accidents per year.

[2]

- (iii) The club management decides that the total of the premiums should be 50% more than the total average cost of accidents per year, and that each group should pay an amount proportional to its contribution to the total average cost.

Calculate the premium for a member of group S.

[2]

1(i)	Group	P	Q	R	S
	Number of people	15			
	Average number of accidents in a year		4.5		
	Average cost of accidents per year	£7500	£9000		

1(ii)	
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1(iii)	

6 On lines 94 and 95 it says “Table 4 gives the timings for round trips in which the cars are required to stop at every floor they serve; Table 2 suggests this is a common occurrence in this case”. Explain how Table 2 is used to make this claim.

[3]

7	

7 8 people make independent visits to any one of the upper floors of a building with 10 upper floors. What is the probability that at least one of the visitors goes to the top floor?

[2]

6	

(i) Referring to Strategy 3 and lines 99 to 101, complete the table below for car C.

[3]

(ii) Calculate the time car C will take to transport all the people who work on floors 7 and 8, and return to the ground floor.

[1]

5(i)	
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Car C

	Arrival time (seconds)	Departure time (seconds)
Ground floor	0	20
Floor 1		
Floor 2		
Floor 3		
Floor 4		
Floor 5		
Floor 6		
Floor 7		
Floor 8		
Floor 9		
Floor 10		
Return to ground floor		

5 (ii)



- 9 In lines 89 and 90 it says "... on average there will be approximately 8 stops per trip. A round trip with 8 stops would take between 188 and 200 seconds". Explain how the figure of 188 seconds has been derived.

[2]

4	

- 10 Using the expression on line 51, work out the answer to the question on lines 39 and 40 for the case where there are 10 upper floors and 7 people. Give your answer to 2 decimal places.

[2]

3	

11 In line 79 it says “For most journeys, more than half the journey time is composed of load time and transfer time”. For what percentage of the journey time for the round trip made by car A in Table 4 is the car stationary?

[2]

2	

- 12 In a building with a single lift, the car is located on the ground floor. The following calls arrive in quick succession, all before the car has started on its upward journey.

- Passenger P on floor 9 calls for a down-car (intending to go to floor 1);
- Passenger Q on floor 5 calls for an up-car (intending to go to floor 10);
- Passenger R on floor 4 calls for a down-car (intending to go to floor 1).

The following events need to take place, but not necessarily in the order given.

- E1. Pick up P
- E2. Pick up Q
- E3. Pick up R
- E4. Drop off P and R
- E5. Drop off Q

Assuming the car serves requests as described in lines 16 to 20,

(i) which event occurs first?

[1]

(ii) which event occurs second?

[1]

(iii) which event occurs last?

[1]

1 (i)	
1 (ii)	
1 (iii)	

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